

Oops, I add it again...

Error in aggregating futures return

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Abstract

The futures price series F_t is sampled from adjoining contracts that trade at different prices on roll-days. As such, F_t is periodically discontinuous which, if left unaddressed, will lace the return series with jump error. To address this issue, data vendors and experienced researchers would adjust F_t to ensure that returns are derived from same-contract prices. But doing so implies a derived return series that correspond to different contract positions, which cannot be directly aggregated over time. Doing so will cause the cumulative futures return to be a function of the average term slope observed on roll-days, which resembles a *roll yield*. Based on 12 commodity futures over 30 years, we show that the errors are non-trivial. We further contribute a solution to aggregate futures return without error, which addresses the roll yield misconception, and prevents misleading evaluation on the investment appeal of commodity futures.

JEL classification: G14, G15.

Keywords: Price jump; Commodity futures; Roll yield.

1 Introduction

Futures markets provide investors easy access to different asset classes, with minimal trading friction and transaction cost. A wide range of investment analysis and portfolio evaluation, utilize return and volatility from the futures prices of different underlying assets. Liquidity generally cluster on shorter term contracts, and in most markets, on the front contract. For back-testing purposes, price data is usually sampled by periodically rolling from the front contract to the next-cycle contract to construct a daily futures price series F_t over a long sample period. The sampled F_t is used to generate a return series R_t , which is widely used across many empirical applications.

Unlike the price series of many financial securities, F_t is periodically discontinuous on every roll-day $T_i, i = 1, 2, \dots, N$ over the sample period; On T_i , the sampling procedure switches between adjoining contracts, causing F_t to jump from the old contract price $F_{T_i}^o$ to the new contract price $F_{T_i}^n$. Since adjoining contracts have different maturities, cost-of-carry implies they trade at different prices $F_{T_i}^o \neq F_{T_i}^n$. As such, F_t exhibits periodic roll-day jumps which, if left unaddressed, will lace R_t with jump error, and expose the empirical analysis to potentially non-trivial biases. The jump error occurs as follows; On roll-day T_i , the old contract price $F_{T_i}^o$ is sampled for the last time. From $T_i + 1$, the sampled price $F_{T_{i+1}}^n$ comes from the new contract. This causes the post roll-day return $R_{T_{i+1}}$ to be *wrongly* calculated using adjoining contract prices $F_{T_{i+1}}^n$ and $F_{T_i}^o$, instead of $F_{T_i}^n$. The jump error is due to $F_{T_i}^o \neq F_{T_i}^n$, which could be interpreted as a *roll yield*, but it does not generate any profit/loss from rolling a futures position over time. Indeed, Bessembinder (2018) rightly emphasizes that no cashflow can occur, and no return can be earned from a price difference between futures contracts.

To address the discontinuity in F_t , Bessembinder (2018) emphasizes on calculating futures return using same-contract prices. Erb and Harvey (2006) and Gorton and Rowenhurst (2006) use the cumulative roll-day price jump between adjoining contracts to adjust commodity prices, so as to derive a return series from same-contract prices. Bloomberg and Standard & Poor's Dow Jones (SPDJ) use a weighted-average sampling approach to construct their commodity price indices; Their approach gradually shifts the weighting from the old to new contract price by 20% a day, over 5 roll days, instead of 100% in a roll-day. This approach smooths out the discontinuity in F_t over a trading week, which allows a return series to be derived from 'almost' same-contract prices.

Our paper highlights a *residual* jump error problem that occurs when we aggregate a return series R_t that is derived from same-contract prices. To the best of our knowledge, this issue has not been addressed in the literature. The cause of the error is subtle, but intuitively obvious; If R_t is derived from same-contract prices, it must correspond to different contract positions over time. But return aggregation assumes an on-going position, whether as the product-sum of discrete return, or linear-sum of log return. Even when a futures position is rolled instantaneously between adjoining contracts, it is still discontinuous, if the position is switched at different prices. This oversight leaves a residual jump error in the cumulative return CR , which we show is a function of term slopes observed on roll-days over

an investment horizon. This implies CR is measured with error; Term slope cannot affect the return from an on-going futures position.

The finding sheds light on a source of the roll yield misconception. Market commentators and academics who are unfamiliar with futures trading, would routinely (and wrongly) interpret roll yield as a return from rolling a futures position along the term slope. If a sampled F_t is periodically discontinuous due to roll-day price jumps, and this is not addressed, roll yield will periodically manifest in R_t as jump error. But if data vendors and experienced researchers already address the issue, shouldn't that clear up the roll yield misconception?

Analytically, we show that aggregating R_t that is free from jump error, also add up term slopes on roll days. The latter is either a measure of roll yield, or at least highly correlated with a roll yield measure in the literature. This suggests that the roll yield misconception still clouds an empirical analysis that aggregates futures return over time. Our results, based on 12 commodity futures markets over 30 years, show that the error is empirically non-trivial for the majority of commodities. As residual jump error accumulates over time, its severity increases with the investment horizon, especially for commodities that are in a pervasively contango or backwardation state.

Our paper complements studies that are aware of the roll-yield misconception in futures trading, including Erb and Harvey (2006, 2016), Sanders and Irwin (2012), Gorton et al. (2013), Bhardwaj, Gorton, and Rouwenhorst (2015), Bessembinder et al. (2016), and Bessembinder (2018). Erb and Harvey (2006) decompose futures return into excess return and roll yield, where excess return corresponds to a continuous price series that is free from roll-day jumps, and is used by Sanders and Irwin (2012), Irwin et al. (2020), and Moran et al. (2020), among others. Gorton and Rouwenhorst (2006) and Gorton et al. (2013) emphasize on calculating futures return using prices from the same contract; This is consistent with the Erb and Harvey (2006) excess return measure. Bessembinder (2018) examines realized return from profit/loss cashflow over time, but did not examine it as a return calculation like Erb and Harvey (2006). These studies focus on addressing roll-yield in return calculation, but not in return aggregation.

We also contribute a solution to aggregate futures return without error. We derive the analytical form of residual jump error, which is a cumulative price gap between adjoining contracts on roll-days. This provides an insight to develop a two-step solution to aggregate futures return without error¹. Analytically, our CR measure is not a function of term slope; It does not contain any component that relates to roll yield. Our proposed aggregation method helps to prevent potentially misleading evaluation on the investment appeal of commodity futures markets.

¹In step 1, we sum up same-contract daily return as per normal, since they correspond to an on-going position. This maps R_t from daily frequency, to a contract-specific cumulative return series CR_{T_i} in contract-cycle frequency T_i , $i = 1, 2, \dots, N$. In step 2, we can either i) linear-sum CR_{T_i} as an approximation, or ii) apply a scaling adjustment on CR_{T_i} , then aggregate as per normal. The methods differ in the normalizing base price, but both return measures focus on the cumulative profit/loss cashflow from contract-specific positions over time.

We evaluate our proposed solution against standard aggregation, using 3 futures price series; The first 2 are commodity price indices from Bloomberg and SPDJ. The third price series comes from sampling prices from adjoining futures contracts, after which we apply a similar adjustment as Erb and Harvey (2006) and Gorton and Rowenhurst (2006) on the sampled F_t to derive the return series. Given the 3 return series are not periodically affected by jump error, any discrepancy between R_z and a given CR measure, can be attributed to the aggregation process itself.

Our proposed aggregation requires the identification of roll days, so it cannot be readily applied on data vendor price indices. This gives 5 cumulative return measures to evaluate; 3 from standard aggregation on the 3 price series, and our two variant CR measures on the sampled F_t series. For a valid evaluation, we need to benchmark the five CR measures against the actual realized return R_z that is earned from the same futures position. To compute R_z , we manually track the daily mark-to-market cumulative profit/loss on the futures position's margin account. We normalize the closing balance by the initial price F_{T_0} that the futures position is entered into, to calculate R_z .

Our results show that standard aggregation produces comparable CR measures among the 3 futures price series. This seems to suggest that we can aggregate futures return in the usual way, and measure CR that is 'consistent' among commonly used source of futures price data. However, we show that the standard measures of CR , differ substantially from our two variant measures of CR on the sampled F_t . More importantly, the 3 standard CR measures deviate substantially from R_z , while our two CR measures are much closer. This suggests that a standard aggregation of returns derived from commonly sourced futures price series, could produce a CR measure that is not what is actually earned from the futures position in terms of profit/loss, over the same investment horizon.

To acquire more insight, we examine how does the discrepancy i.e. residual jump error, behave across different commodities. While residual jump error is an important conceptual oversight in the literature, whether it poses an empirical concern, depends on two market conditions: Persistence in term slope² and market direction³. For example, if a futures market exhibits a persistently positive (negative) term slope, and its price series also exhibits a strong upward (downward) trend, this will cause more severe residual jump errors in CR . We raise awareness on the market conditions under which readers should be more careful when aggregating futures return, so as to avoid misleading interpretation on the profitability of futures positions.

The paper proceeds as follows. The next section outlines the residual jump error problem in detail. Section 3 presents our approach to rectify residual jump errors. Section 4 compares empirical results on CR , with and without residual jump error, for 12 different commodity futures markets. Section 5 concludes.

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²Is the term slope predominantly positive or negative during the investment horizon?

³Is F_t trending upwards or downwards during the investment horizon?

2 The residual jump error problem

To give readers a better sense of how residual jump error occurs during return aggregation, we give a brief outline of the jump error problem in futures return, and how it is addressed in the literature.

2.1 Roll-day price jump and jump error in return

Consider a futures price series that is sampled across 3 contracts $\{F_1, F_2, F_3\}$ over 2 roll days $\{T_1, T_2\}$, where the term slope is not flat $F_{1,T_1} \neq F_{2,T_1}$ and $F_{2,T_2} \neq F_{3,T_2}$. Without loss of generality, assume there are no trading days between roll-days and consider discrete return.

Sampled prices: $\{F_{1,T_0}, F_{1,T_1}, F_{2,T_2}, F_{3,T_3}\}$

Discrete returns: $R_{T_1} = \frac{F_{1,T_1}}{F_{1,T_0}} - 1$, $R_{T_2}^{je} = \frac{F_{2,T_2}}{F_{1,T_1}} - 1$, $R_{T_3}^{je} = \frac{F_{3,T_3}}{F_{2,T_2}} - 1$

$$\begin{aligned}
 CR &= [(1 + R_{T_1})(1 + R_{T_2}^{je})(1 + R_{T_3}^{je})] - 1 = \left(\frac{F_{1,T_1}}{F_{1,T_0}}\right)\left(\frac{F_{2,T_2}}{F_{1,T_1}}\right)\left(\frac{F_{3,T_3}}{F_{2,T_2}}\right) - 1 \\
 &= \begin{cases} \frac{F_{3,T_3} - F_{1,T_0}}{F_{1,T_0}}, & \text{if } F_{1,T_1} = F_{2,T_1} \text{ and } F_{2,T_2} = F_{3,T_2} \text{ i.e. no jump error;} \\ \left[\frac{(F_{3,T_3} - F_{3,T_2}) + (F_{2,T_2} - F_{2,T_1}) + (F_{1,T_1} - F_{1,T_0})}{F_{1,T_0}} + \frac{(F_{3,T_2} - F_{2,T_2}) + (F_{2,T_1} - F_{1,T_1})}{F_{1,T_0}} \right] & \\ \text{if } F_{1,T_1} \neq F_{2,T_1} \text{ and } F_{2,T_2} \neq F_{3,T_2} \end{cases} = R_z + \text{Jump error}, \tag{1}
 \end{aligned}$$

In equation (1), if we ignore roll-day price jumps at T_1 and T_2 , and compute returns directly from the sampled price series, $R_{T_2}^{je}$ will be wrongly calculated from adjoining contract prices F_{2,T_2} and F_{1,T_1} . Similarly, $R_{T_3}^{je}$ is wrongly derived from F_{3,T_3} and F_{2,T_2} . Both returns contain jump error; No actual cashflow occurs, hence no return could be earned from a price difference between futures contracts. More importantly, equation (1) also shows that the cumulative return CR takes a familiar form $CR = \left[\frac{F_{3,T_3} - F_{1,T_0}}{F_{1,T_0}} \right]$; Without roll-day price jump, all intermediate prices are irrelevant to an on-going position. The cumulative return CR is simply the difference between the price F_{1,T_0} that the position is opened at T_0 , and the price F_{3,T_3} at which the position is closed at T_3 .

But if return contains jump error, then equation (1) shows that CR has two terms; The first is a realized return R_z that is earned from the profit/loss cashflow that is aggregated

across the three contract positions that are closed at T_1, T_2 , and T_3 , normalized by the initial price F_{1,T_0} that the futures position was opened. Put simply, R_z measures what is actually earned from the cumulative profit/loss that flows through a futures trader's margin account. The second term is the jump error, which captures the roll yield that is 'earned' from the roll-day price gap between adjoining contracts.

2.2 Adjusted futures prices to address jump error

We outline a similar method approach that Erb and Harvey (2006) and Gorton and Rowen-hurst (2006) use to address jump error in futures return. The approach uses the price gap between adjoining contracts on roll-days T_i , as a jump-ratio $JR_{T_i} = \frac{F_{i,T_i}}{F_{i+1,T_i}}$ to adjust subsequent prices that are sampled from the newly-rolled contract. The purpose of JR_{T_i} is to offset the price jump, and form a continuous price series across adjoining contracts. The jump-ratio is cumulative; Prices from the third contract are adjusted using the product of jump ratios over the two roll-days, and so forth. We outline the jump-ratio (JR) method in equation (2), based on the same setting as equation (1).

Sampled prices: $\{F_{1,T_0}, F_{1,T_1}, F_{2,T_2}, F_{3,T_3}\}$

JR prices: $F_{1,T_0}, F_{1,T_1}, F_{2,T_2}^{jr} = \left(\frac{F_{1,T_1}}{F_{2,T_1}}\right) F_{2,T_2}, F_{3,T_3}^{jr} = \left(\frac{F_{1,T_1}}{F_{2,T_1}}\right) \left(\frac{F_{2,T_2}}{F_{3,T_2}}\right) F_{3,T_3}$

JR returns: $R_{1,T_1} = \frac{F_{1,T_1}}{F_{1,T_0}} - 1, R_{jr,T_2} = \frac{F_{2,T_2}^{jr}}{F_{1,T_1}} - 1 = \frac{F_{2,T_2}}{F_{2,T_1}} - 1 = R_{2,T_2}$

$R_{jr,T_3} = \frac{F_{3,T_3}^{jr}}{F_{2,T_2}^{jr}} - 1 = \frac{F_{3,T_3}}{F_{3,T_2}} - 1 = R_{3,T_3}$

$CR_{jr} = \left(\frac{F_{1,T_1}}{F_{1,T_0}}\right) \left(\frac{F_{2,T_2}}{F_{2,T_1}}\right) \left(\frac{F_{3,T_3}}{F_{3,T_2}}\right) - 1 = \left(\frac{F_{1,T_1}}{F_{2,T_1}}\right) \left(\frac{F_{2,T_2}}{F_{3,T_2}}\right) \frac{F_{3,T_3}}{F_{1,T_0}} - 1 = \left(\frac{F_{1,T_1}}{F_{2,T_1}}\right) \left(\frac{F_{2,T_2}}{F_{3,T_2}}\right) (1 + CR) - 1$

$$(1 + CR_{jr}) = \left(\frac{F_{1,T_1}}{F_{2,T_1}}\right) \left(\frac{F_{2,T_2}}{F_{3,T_2}}\right) \left[1 + R_z + \frac{(F_{3,T_2} - F_{2,T_2}) + (F_{2,T_1} - F_{1,T_1})}{F_{1,T_0}}\right] \quad (2)$$

At time T_2 , the JR return $R_{jr,T_2} = R_{2,T_2}$ is computed from prices of the second rolled contract; Unlike $R_{T_2}^{je}$ in equation (1), it does not contain jump error. The same applies to $R_{jr,T_3} = R_{3,T_3}$ at time T_3 . As such, the return series $\{R_{1,T_1}, R_{2,T_2}, R_{3,T_3}\}$ is derived from same-contract prices, and is free from jump error. However, this also implies the return

series, correspond to different contract positions. At T_1 , the position is rolled from F_{1,T_1} to $F_{2,T_1} \neq F_{1,T_1}$. As such, the position is not ongoing, even if the switch occurs instantaneously. The same applies at T_2 , where the position is switched from F_{2,T_2} to F_{3,T_2} . Return aggregation assumes an ongoing position, which equation (2) shows will leave a residual jump error in the cumulative JR return CR_{jr} . Only when there are no roll-day price jumps, would $CR_{jr} = R_z$. This shows that while the JR return series is free from jump error, its aggregation will leave a residual jump error in CR .

Equation (2) also provides a technical insight into the source of the residual jump error. In equation (1), the jump error takes the form $\left[\frac{(F_{3,T_2} - F_{2,T_2}) + (F_{2,T_1} - F_{1,T_1})}{F_{1,T_0}} \right]$. Equation (2) shows that the JR method applies a cumulative jump-ratio $(\frac{F_{1,T_1}}{F_{2,T_1}})(\frac{F_{2,T_2}}{F_{3,T_2}})$ to rectify the error. Consider a futures price series that is sampled on a positive term slope; Price up-jumps induce a positive up-jump error in CR , which is offset by a greater discounting via $(\frac{F_{1,T_1}}{F_{2,T_1}})(\frac{F_{2,T_2}}{F_{3,T_2}}) < 1$. Similarly, sampling from a negative term slope causes negative down-jump error in CR , which is offset by a greater compounding via $(\frac{F_{1,T_1}}{F_{2,T_1}})(\frac{F_{2,T_2}}{F_{3,T_2}}) > 1$. In either scenario, CR_{jr} contains an offsetting adjustment, such that it will always be closer (than CR) to the actual realized return R_z . However, the adjustment is imprecise; The jump error in CR manifests at the price level, but the JR method applies a proportional adjustment. This leaves a residual jump error in CR_{jr} which, like jump error, is also a function of the cumulative price gap between adjoining contracts on roll-days over the investment horizon.

2.3 An aggregation solution

We outline a two-step solution to aggregate a futures return series without residual jump error. In step 1, we directly aggregate same-contract daily returns $R_{i,t}$, which correspond to an on-going position. This maps a daily futures return series $R_{i,t}$, to a contract-specific cumulative return series CR_{T_i} , which is in contract cycle frequency T_i .

Step 1: Aggregate same-contract returns

$$CR_{T_i} = \prod_{t=1}^{T_i} (1 + R_{i,t}) - 1, \text{ for each contract } i = 1, 2, \dots, N$$

Cumulative return series: $\{CR_{T_1}, CR_{T_2}, \dots, CR_{T_N}\}$

In Step 2, we aggregate the CR_{T_i} series over the investment horizon. There are two variant methods 2A and 2B; Step 2A is an approximation that treats each contract position separately i.e. no compounding, and takes a linear sum of the CR_{T_i} series over N roll-days i.e. $\sum_i^N CR_{T_i}$. It does not explicit account for the profit/loss from one position, being carried

into the next position, and is likely to contain an approximation error. Empirically, we are interested in comparing the potential error in Step 2A, against residual jump error.

Equation (3) shows that step 2A generates a cumulative return measure CR_A that normalizes the total profit/loss cashflow CF_{i,T_i} from each contract i , by the price that each contract was rolled into $F_{i,T_{i-1}}$. Put simply, CR_A is the sum of contract-specific realized return $R_{z,i}$ for $i = 1, 2, \dots, N$. Taking CR_A/N gives an average realized return μ_{R_z} from holding a given commodity futures position, for 1 contract cycle.

Step 2A: Linear-sum CR_{T_i} across contracts

$$\begin{aligned}
CR_A &= \sum_i^N CR_{T_i} = \left(\frac{F_{1,T_1} - F_{1,T_0}}{F_{1,T_0}} \right) + \left(\frac{F_{2,T_2} - F_{2,T_1}}{F_{2,T_1}} \right) + \dots + \left(\frac{F_{N,T_N} - F_{N,T_{N-1}}}{F_{N,T_{N-1}}} \right) \\
&= \left[\frac{CF_{1,T_1}}{F_{1,T_0}} + \frac{CF_{2,T_2}}{F_{2,T_1}} + \dots + \frac{CF_{N,T_N}}{F_{N,T_{N-1}}} \right] = \sum_i^N \frac{CF_{i,T_i}}{F_{i,T_{i-1}}} = \sum_i^N R_{z,i} \\
\sum_i^N R_{z,i}/N &= \mu_{R_z}
\end{aligned} \tag{3}$$

Next, we outline step 2B, which accurately captures the return process that is associated with rolling a futures position over contracts, albeit more complex. After step 1, an adjustment is required on the series CR_{T_i} , after which the adjusted series aCR_{T_i} can be aggregated without error, to compute CR_B over the investment horizon. Equation (4) outlines step 2B, where $J_{T_i} = F_{i,T_i} - F_{i+1,T_i}$ is the roll-day price gap between adjoining contracts.

Step 2B: Product-sum adjusted CR_{T_i} across contracts

$$\begin{aligned}
aCR_{T_1} &= \frac{F_{1,T_1}}{F_{1,T_0}} - 1 = CR_{T_1}, \quad aCR_{T_2} = \frac{F_{2,T_1}}{F_{1,T_1}}(CR_{T_2}), \\
aCR_{T_3} &= \frac{F_{3,T_2}}{F_{2,T_2} + [F_{1,T_1} - F_{2,T_1}]}(CR_{T_3}) = \frac{F_{3,T_2}}{F_{2,T_2} + J_{T_1}}(CR_{T_3}), \\
aCR_{T_4} &= \frac{F_{4,T_3}}{F_{3,T_3} + [J_{T_1} + J_{T_2}]}(CR_{T_4}), \dots, aCR_{T_N} = \frac{F_{N,T_{N-1}}}{F_{N-1,T_{N-1}} + \sum_{i=1}^{N-2} J_{T_i}}(CR_{T_N})
\end{aligned}$$

$$CR_B = \prod_i^N (1 + aCR_{T_i}) - 1 = \frac{\sum_i^N CF_{i,T_i}}{F_{1,T_0}} = R_z \quad (4)$$

Both CR_A and CR_B are similar in that they focus on aggregate profit/loss cashflow from each futures position over time. The main difference lies in the normalizing base price; CR_A uses each contract position's entry price, while CR_B uses the price that sets up the initial futures position at time T_0 , to normalize total profit/loss cashflow over the investment horizon $T_0 \sim T_N$. As such, $CR_A/N = \mu_{R_z}$ gives a short-horizon measure of average realized return over a contract-cycle, while CR_B is a longer-horizon measure of realized return from rolling a futures position over multiple contract cycles $i = 1, 2, \dots, N$. To note, CR_B has the same functional form as our definition of realized return R_z in equation (1).

Some readers may query how we come up with the adjustment in step 2B, or whether $CR_B = R_z$ actually holds. We address both issues below.

2.3.1 The intuition behind step 2B

We outline the intuition that guides step 2B. Step 1 generates a CR_{T_i} series that corresponds to separate contract positions. As such, it is incompatible with standard aggregation, which assumes an on-going position. The aim of step 2B is to scale CR_{T_i} so that it resonates with an on-going position. This would allow the initial futures price F_{1,T_0} to ‘travel continuously’ from the first contract to the second F_2 , and onto the third contract F_3 , until it reaches the final price F_{N,T_N} at which point the futures position is closed out at time T_N .

We start with F_{1,T_0} , where CR_{T_1} describes its ‘journey’ to become F_{1,T_1} . But at T_1 , it is F_{2,T_1} , and not F_{1,T_1} , that continues the journey to F_{2,T_2} , which is described by CR_{T_2} . The scaling ($\frac{F_{2,T_1}}{F_{1,T_1}}$) on CR_{T_2} , allows F_{1,T_1} to ‘pass the baton’ to F_{2,T_1} , thereby connecting the two journeys CR_{T_1} and CR_{T_2} . At T_2 , a similar scaling is required for F_{2,T_2} to ‘pass the baton’ to F_{3,T_2} . But the scaling also requires an extra term $J_{T_1} = F_{1,T_1} - F_{2,T_1}$ in the denominator i.e. ($\frac{F_{3,T_2}}{F_{2,T_2} + J_{T_1}}$); The term J_{T_1} takes into account that at T_1 , CR_{T_2} has been adjusted by F_{1,T_1} and F_{2,T_1} . This has to be recognized by CR_{T_3} at T_2 , so that it can connect with CR_{T_2} . Doing so allows F_{3,T_2} to continue the journey described by CR_{T_3} , to reach F_{3,T_3} at T_3 , and so forth.

In sum, the adjusted series aCR_{T_i} corresponds to an on-going futures position over contract-cycles, which is entered into at the start of the investment horizon T_0 . This allows us to aggregate aCR_{T_i} directly to compute CR_B , which measures the total profit/loss cashflow over the investment horizon, normalized by the initial futures price F_{1,T_0} .

2.3.2 The proof for step 2B: $CR_B = R_z$

We provide a simplified proof that aggregating aCR_{T_i} , gives $CR_B = R_z$. Without loss of generality, consider 3 contract cycles over 2 roll-days. We can confirm the same result holds for a longer series aCR_{T_N} over N contract cycles.

$$\begin{aligned}
CR_B &= \prod_i^3 = (1 + aCR_{T_i}) - 1 = \left(\frac{F_{1,T_1}}{F_{1,T_0}}\right)\left(1 + \frac{F_{2,T_1}}{F_{1,T_1}}CR_{T_2}\right)\left(1 + \frac{F_{3,T_2}}{F_{2,T_2} + J_{T_1}}CR_{T_3}\right) - 1 \\
&= \left(\frac{F_{1,T_1}}{F_{1,T_0}}\right)\left(\frac{F_{1,T_1} + F_{2,T_1}CR_{T_2}}{F_{1,T_1}}\right)\left(\frac{F_{2,T_2} + J_{T_1} + F_{3,T_2}CR_{T_3}}{F_{2,T_2} + J_{T_1}}\right) - 1 \\
&= \left(\frac{F_{1,T_1}}{F_{1,T_0}}\right)\left(\frac{F_{1,T_1} + F_{2,T_2} - F_{2,T_1}}{F_{1,T_1}}\right)\left(\frac{F_{2,T_2} + F_{1,T_1} - F_{2,T_1} + F_{3,T_3} - F_{3,T_2}}{F_{1,T_1} + F_{2,T_2} - F_{2,T_1}}\right) - 1 \\
&= \left(\frac{F_{3,T_3} - F_{3,T_2} + F_{2,T_2} - F_{2,T_1} + F_{1,T_1}}{F_{1,T_0}}\right) - 1 \\
&= \left(\frac{CF_{3,T_3} + CF_{2,T_2} + CF_{1,T_1}}{F_{1,T_0}}\right) = \frac{\sum_i^3 CF_{i,T_i}}{F_{1,T_0}} = R_z
\end{aligned}$$

The above shows that CR_B measures cumulative return as the total profit/loss cashflow over contracts, normalized by the initial futures price F_{1,T_0} that the position was first entered into. The choice between 2A and 2B evokes a trade-off between simplicity and flexibility; Step 2A is simple to use, but it measures the average realized return over 1 contract cycle. Step 2B accurately measures the realized return from a futures position, for any investment horizon, albeit more complex to apply. To note, unlike CR or CR_{jr} , both CR_A and CR_B are not functions of the term slope.

2.4 Measuring residual jump error

Analytically, we have shown that directly aggregating a futures return series, will leave a residual jump error in the cumulative return CR_{jr} . But empirically, how serious is the problem? What are the market conditions under which analysts and academics need to be careful when working with aggregate futures return? To shed light on these questions, we derive the analytical form for residual jump error ε_{T_N} to gain some insight on how it behaves over time, and across different markets. This allows us to identify the market conditions that affect the severity of ε_{T_N} .

If we directly aggregate a daily futures return series, this is equivalent to using step 1 to generate the CR_{T_i} series, and repeating step 1 to aggregate CR_{T_i} into CR_{jr} . Below, we show

that CR_{jr} is an incoherent mix of prices across the 3 contract positions. Equation (2) also shows this, where the jump-ratio term does not cancel against the jump-error term precisely.

$$\begin{aligned} CR_{jr} &= \prod_i^3 (1 + CR_{T_i}) - 1 = \left(\frac{F_{1,T_1}}{F_{1,T_0}}\right)\left(\frac{F_{2,T_2}}{F_{2,T_1}}\right)\left(\frac{F_{3,T_3}}{F_{3,T_2}}\right) - 1 \\ &= \left(\frac{F_{1,T_1}F_{2,T_2}F_{3,T_3} - F_{1,T_0}F_{2,T_1}F_{3,T_2}}{F_{1,T_0}F_{2,T_1}F_{3,T_2}}\right) \end{aligned}$$

In equation (5), we generalize the above to cover any investment horizon T_N . Denote $CJR_{T_i} = \prod_i^{N-1} \left(\frac{F_{i,T_i}}{F_{i+1,T_i}}\right)$ as a cumulative jump-ratio, and $JE_{T_i} = F_{i+1,T_i} - F_{i,T_i}$ as the price gap between adjoining contracts observed on roll day T_i . Consider a company's stock, 10-year Treasury bond, gold bar, or any security that allows an on-going position; With no roll-day price jumps, equation (5) shows that $\sum_i^N JE_{T_i} = 0$, $CJR_{T_i} = 1$, and $CR_{jr} = R_z$.

$$\begin{aligned} CR_{jr} &= \prod_i^N (1 + CR_{T_i}) - 1 = \left(\frac{F_{1,T_1}}{F_{1,T_0}}\right)\left(\frac{F_{2,T_2}}{F_{2,T_1}}\right)\dots\left(\frac{F_{N,T_N}}{F_{N,T_{N-1}}}\right) - 1 \\ &= \left(\frac{F_{1,T_1}}{F_{2,T_1}}\right)\left(\frac{F_{2,T_2}}{F_{3,T_2}}\right)\dots\left(\frac{F_{N-1,T_{N-1}}}{F_{N,T_{N-1}}}\right) \times \left(\frac{F_{N,T_N}}{F_{1,T_0}}\right) - 1 \\ &= CJR_{T_i} \left(\frac{F_{N,T_N}}{F_{1,T_0}}\right) - 1, \text{ where } CJR_{T_i} = \prod_i^{N-1} \left(\frac{F_{i,T_i}}{F_{i+1,T_i}}\right) \\ (1 + CR_{jr}) &= CJR_{T_i} \left(1 + R_z + \frac{\sum_i^N JE_{T_i}}{F_{1,T_0}}\right), \text{ where } JE_{T_i} = F_{i+1,T_i} - F_{i,T_i} \end{aligned} \quad (5)$$

In equation (6), we derive residual jump error as the difference between the actual realized return and cumulative return $\varepsilon_{T_N} = R_z - CR_{jr}$, which reveals two components. The first is a cashflow error $[1 - CJR_{T_i}](1 + R_z)$; It shows that R_z , which measures total profit/loss cashflow, is scaled by the adjustment ratio CJR_{T_i} . Data vendors and prior studies adjust futures prices to rectify roll-day price jumps, which allows a return series to be derived without jump error. However, the adjustment is also applied on settlement prices observed on non-roll days. As such, daily change in adjusted prices, do not correspond to daily mark-to-market profit/loss cashflow that a futures position generates. The cashflow error hints at a source of the roll-yield misconception i.e. that a long futures position earns a negative (positive) yield if it is consistently being rolled on a positive (negative) term slope⁴; Even if a return series does

⁴When the term slope is positive, maintaining a long position requires periodically closing expiring posi-

not contain roll-yield, aggregating the series will wrongly involve calculations using prices between adjoining contracts, which relate back to a roll-yield measure.

The second term is an adjustment error; The jump error in futures return manifests at the price level $\sum_i^N JE_{T_i}$. But CJR_{T_i} is a proportional adjustment that allow returns to be derived from same-contract prices. Larger up-jumps on roll-day contribute to a larger $\sum JE_{T_i}$. But CJR_{T_i} also imposes a heavier discounting on $\sum JE_{T_i}$. While there is offsetting, it is imprecision, which manifest as an adjustment error in ε_{T_N} . To note, without any roll-day price jump, $CJR_{T_i} = 1$, $\sum JE_{T_i} = 0$ and equation (6) reduces to $\varepsilon_{T_N} = 0$, which is consistent with equation (5) showing $CR_{jr} = R_z$.

$$\begin{aligned}\varepsilon_{T_N} &= R_z - CR_{jr} = R_z - CJR_{T_i} \left(1 + R_z + \frac{\sum_i^N JE_{T_i}}{F_{1,T_0}} \right) + 1 \\ &= [1 - CJR_{T_i}](1 + R_z) - CJR_{T_i} \left[\frac{\sum_i^N JE_{T_i}}{F_{1,T_0}} \right]\end{aligned}\quad (6)$$

The analytical form of ε_{T_N} reveals how residual jump error could behave, over time and across futures markets. The first term is an interaction between CJR_{T_i} and R_z . This suggests that ε_{T_N} accumulates over time, such that its magnitude could increase with the investment horizon. But the sign of ε_{T_N} depends on two market conditions: i) Stability of the term slope, which affects CJR_{T_i} ; ii) Market direction, which relates to R_z . Both conditions vary over time and across futures markets.

Assume a long futures position, and consider a stable positive term slope i.e. $CJR_{T_i} < 1$ and $\sum_i^N JE_{T_i} > 0$. Given the latter, the adjustment error is always negative. But the sign of ε_{T_N} also depends on $[1 - CJR_{T_i}](1 + R_z)$, which depends on whether the market direction is up $R_z > 0$, down $-1 < R_z < 0$, or severely down $R_z < -1$. As long as margin calls are met, it is possible for the cumulative loss to exceed the initial futures price F_{1,T_0} . We elaborate on three scenarios below.

1. Upmarket $R_z > 0$, such that $[1 - CJR_{T_i}](1 + R_z) > 0$. Given the cashflow error has a compound factor $(1 + R_z) > 1$, while the adjustment error has a discount factor $CJR_{T_i} < 1$, it is likely that $\varepsilon_{T_N} > 0$.
2. Downmarket $-1 < R_z < 0$. When R_z is negative, cashflow error becomes smaller, albeit still positive. But now, it is likely the adjustment error dominates, such that

tions by selling at lower prices, and buying longer-dated contracts at higher prices to re-open the position; When rolling on a negative term slope, a trader periodically sells expiring contracts at higher prices, and re-purchase longer-dated contracts at lower prices, thereby ‘earning’ a positive yield. Bessembinder (2018) explains that this is a misconception; No cashflow or return can be earned from a difference in price between futures contracts.

$\varepsilon_{T_N} < 0$. In the next section, we run some numerical simulations, which confirm that this is the case over a range of values for R_z and CJR_{T_i} .

3. Analytically, for a severe downmarket $R_z \leq -1$, the cashflow error $[1 - CJR_{T_i}](1 + R_z) \leq 0$ has the same sign as the adjustment error, such that $\varepsilon_{T_N} < 0$.

During an upmarket $R_z > 0$, a positive residual jump error $\varepsilon_{T_N} > 0$ corresponds to CR_{jr} under-estimating the actual realized profit from a long futures position, based on $\varepsilon_{T_N} = R_z - CR_{jr}$. To follow, during a downmarket $R_z < 0$, CR_{jr} also under-estimates the actual realized loss, given $\varepsilon_{T_N} < 0$. In sum, when a futures market exhibits a predominately positive term slope during the investment horizon, CR_{jr} is likely to under-estimate the magnitude of R_z , regardless of its sign. In the next section, our simulation results confirm that this is the case.

Now, we outline corresponding scenarios for ε_{T_N} under a stable negative term slope $CJR_{T_i} > 1$ and $\sum_i^N JE_{T_i} < 0$. Here, the adjustment error is always positive.

1. Upmarket $R_z > 0$, such that $[1 - CJR_{T_i}](1 + R_z) < 0$. Given $(1 + R_z) > 1$, the cashflow error magnitude is likely to exceed the adjustment error, such that $\varepsilon_{T_N} < 0$.
2. Downmarket $-1 < R_z < 0$. Here, $[1 - CJR_{T_i}](1 + R_z) < 0$ still holds, but with a smaller magnitude than the adjustment error. As such, we expect $\varepsilon_{T_N} > 0$.
3. Analytically, when $R_z \leq -1$, this will lead to $[1 - CJR_{T_i}](1 + R_z) \geq 0$. Since both error terms are positive, this guarantees $\varepsilon_{T_N} > 0$.

During an upmarket $R_z > 0$, $\varepsilon_{T_N} < 0$ corresponds to CR_{jr} over-estimating realized profit. And for a downmarket $R_z < 0$, CR_{jr} also over-estimates the realized loss, such that $\varepsilon_{T_N} > 0$. Hence under a stable negative term slope, CR_{jr} has a tendency to over-estimate the magnitude of R_z , whether the futures position makes a profit, or incurs a loss.

The discussion reveals that CR_{jr} underestimates (overestimates) the magnitude of realized return under a positive (negative) term slope, for both profit and loss positions. To address jump error in futures return, the price series is adjusted using CJR_{T_i} . When the term slope is mostly positive, $CJR_{T_i} < 1$ applies a discounting adjustment to pull down a price series that has been inflated by roll-day up-jumps. But the discounting is also applied to prices on non-roll days. This causes the cumulative return CR_{jr} from a discounted price series to have a smaller magnitude than R_z , regardless of whether the price series is trending up or down. Conversely, when the term slope is mostly negative, $CJR_{T_i} > 1$ delivers a compounding adjustment on a price series that has been deflated by roll-day down-jumps. But this also applies to prices on non-roll days, which amplify both positive and negative returns, causing CR_{jr} to overstate the magnitude of R_z , in both market directions.

Based on the above, we make several empirical predictions on the severity of residual jump error that arises from directly aggregating futures return CR_{jr} , under different market

conditions. For a given investment horizon, if a futures position is held under a term slope that is predominately:

1. **Positive:** CR_{jr} under-estimates the magnitude of actual realized return R_z , whether the futures position earns a profit or suffers a loss.
2. **Negative:** CR_{jr} over-estimates the magnitude of R_z , whether the futures position earns a profit or suffers a loss.
3. **Flat:** This includes both a term slope that is literally flat, or one that is cumulatively flat towards the end of the investment horizon i.e. when positive and negative term slopes are observed during the investment horizon. Here, CJR_{T_i} is close to 1, $\sum_i^N JE_{T_i}$ is close to zero, such that ε_{T_N} is not a serious concern empirically.

Equation (6) further reveals the asymmetric nature of residual jump error. Assume losses are limited to $R_z > -1$; Whether the cumulative term slope is positive or negative, both error terms in ε_{T_N} have opposite signs, so they offset each other⁵. However, market direction R_z affects the cashflow error, but not the adjustment error. When R_z goes from positive to negative, the cashflow error magnitude reduces, such that adjustment error dominates ε_{T_N} . When $R_z = -1$, ε_{T_N} only has adjustment error. And when there is a more severe downmarket i.e. $R_z < -1$, both error terms have the same sign. This suggests that ε_{T_N} is more concerning during downmarket.

While this holds irrespective of the sign of the cumulative term slope, equation (6) also shows that ε_{T_N} is amplified when a downmarket coincides with a cumulatively negative term slope. In the cashflow error, both components are discount factors i.e. $[1 - CJR_{T_i}] < 1$ and $(1 + R_z) < 1$, while the adjustment error has a compound factor $CJR_{T_i} > 1$. Under negative term slope, CR_{jr} is likely to overstate losses from a downmarket, more severely than it overstates profits from an upmarket. We confirm this using numerical simulation.

3 Empirical analysis on residual jump error

We ascertain the presence of residual jump error in cumulative futures return, and analyze how it varies across commodities under different market conditions. This allows us to verify the predictions that are outline in the previous section, thereby empirically validating the functional form of ε_{T_N} in equation (6).

⁵For example, consider $R_z > 0$ and a positive slope $CJR_{T_i} < 1$, which corresponds to a positive cashflow error and negative adjustment error. The same applies when the cumulative term slope is negative i.e. $CJR_{T_i} > 1$, which gives a negative cashflow error and positive adjustment error.

3.1 Numerical example

We provide a comprehensive numerical example to give readers a better sense on how term slope and market direction affect the two components of residual jump error. To see how residual jump error manifests under different market conditions, we examine 4 scenarios that interact between term slope (positive vs. negative) and market direction (up vs. down). Table 1 presents the sampled and adjusted futures prices, daily unadjusted return and CR , daily adjusted return and CR_{jr} , and R_z , under a positive term slope in panel A. We present a similar set of results in panel B, under a negative term slope. The example is based on the following trading information.

- The trader holds a long futures position over 3 contract cycles $\{F_1, F_2, F_3\}$ across 2 roll-days.
- The initial futures position is opened at \$10, which applies to all scenarios.
- On each of the two roll-days, the price jump is \$0.50, for both positive and negative term slopes. The magnitude of the 2 jumps represent 10% of the initial futures price.
- Actual profit or loss from each contract position, is \$0.50. Hence, total profit or loss is \$1.50, such that the realized return R_z is either 15% during an upmarket, or -15% during a downmarket.
- In panel A, the term slope is cumulatively positive.
 1. **Upmarket: $R_z=15\%$.**
 - Sampled $F_t=\{10, 10.50, 11.50, 12.50\}$, which gives $CR=25\%$.
 - Adjusted $F_t=\{10, 10.50, 10.98, 11.43\}$, which gives $CR_{jr}=14.35\%$.
 2. **Downmarket: $R_z=-15\%$.**
 - Sampled $F_t=\{10, 9.50, 9.50, 9.50\}$, which gives $CR=-5\%$.
 - Adjusted $F_t=\{10, 10.50, 9.03, 8.57\}$, which gives $CR_{jr}=-14.26\%$.
- In panel B, the term slope is cumulatively negative.
 1. **Upmarket: $R_z=15\%$.**
 - Sampled $F_t=\{10, 10.50, 10.50, 10.50\}$, which gives $CR=5\%$.
 - Adjusted $F_t=\{10, 10.50, 11.03, 11.58\}$, which gives $CR_{jr}=15.76\%$.
 2. **Downmarket: $R_z=-15\%$.**
 - Sampled $F_t=\{10, 9.50, 8.50, 7.50\}$, which gives $CR=-25\%$.
 - Adjusted $F_t=\{10, 9.50, 8.97, 8.41\}$, which gives $CR_{jr}=-15.89\%$.

INSERT TABLE 1 HERE

Panel A shows that if returns are derived from the sampled futures price series, they contain up-jump error. As such, CR will overstate R_z by 10% in both upmarket (25% vs. 15%) as well as downmarket (-5% vs. -15%) states. The positive bias in CR comes from the two roll-day up-jumps, which represent 10% of the initial futures price. The adjusted F_t is pulled down by $CJR_{T_i} < 1$, which rectifies the up-jump error in returns. But upon aggregation, $CR_{jr} = 14.35\% < R_z$ and $\varepsilon_{T_N} > 0$ i.e. CR_{jr} understates realized gain in the upmarket state. And in the downmarket state, $CR_{jr} = -14.26\% > R_z$, such that $\varepsilon_{T_N} < 0$ i.e. understates realized loss during downmarket. The numerical results are consistent with equation (6) in suggesting that CR_{jr} under-estimates the magnitude of R_z , regardless of market direction.

Panel B presents a similar set of results, under a negative term slope. The sampled futures prices generate a return series with down-jump errors. This causes CR to understate R_z by 10%, both in the upmarket state (5% vs. 15%) and downmarket state (-25% vs. -15%). Now, the adjusted F_t is pushed up by $CJR_{T_i} > 1$ to rectify down-jump error in returns. But upon aggregation, $CR_{jr} = 15.76\%$ for the upmarket state, and -15.89% for the downmarket state. This is also consistent with our empirical proposition that CR_{jr} over-estimates the magnitude of R_z , regardless of its sign.

In order to tease out the asymmetric nature of ε_{T_N} , we run a simulation on the numerical example by considering a wide range of realized return R_z , and the corresponding values for residual jump error ε_{T_N} . As R_z tends to 100%, this represents an upmarket direction; And when it tends to -100%, this corresponds to a downmarket direction. We run the simulation separately for a positive and negative term slope with a 5% jump on each of 2 roll days. To visualize the asymmetry, we plot ε_{T_N} against R_z in Figure X, where the blue (green) graph corresponds to a positive (negative) term slope. Accordingly, Figure X illustrates how ε_{T_N} behaves under all 4 market conditions interacting up/down market with positive/negative term slope.

There are two aspects of asymmetry. The first aspect compares the left and right side of Figure X. During upmarket (downmarket), $|\varepsilon_{T_N}|$ increases with R_z , but at a decreasing (increasing) rate. This applies whether term slope is positive or negative. Equation (6) predicts this; When $R_z > 0$, the two errors terms in ε_{T_N} offset each other, such that ε_{T_N} increases at a decreasing rate. But during downmarket, a decreasing R_z reduces the magnitude of the cashflow error, such that the adjustment error starts to dominate ε_{T_N} . The second aspect compares the top and bottom graph on the downmarket side; ε_{T_N} is more severe under negative term slope; When $R_z = -90\%$, the residual jump error is 10% when term slope is negative, and -6.8% when it is positive. This implies that ε_{T_N} becomes a concern when a downmarket coincides with a negative term slope.

We notice that both graphs pass the point of origin i.e. when $R_z = 0$, then $\varepsilon_{T_N} = 0$. But according to equation (6), this is not the case. The graphs are based on a simplistic

example, where $R_z = 0$ refers to every contract position having zero profit/loss. The latter guarantees $R_z = 0$, but $R_z = 0$ can also come from many possible profit/loss combinations across positions. The greater the volatility in profit/loss over time, the greater the ε_{T_N} , but proofing this is beyond the scope of the current paper.

As a robustness check, we switch the horizontal axis from R_z to cumulative term slope, which we measure as the sum of roll-day jumps divided by initial futures price i.e. a proxy of roll-yield. We present a pair of graphs that represent an upmarket (blue) and downmarket (green) state, corresponding to $R_z = \pm 10\%$. In Figure Y, the right (left) side refers to a positive (negative) slope, while the two graphs represents up and down market, such that Fig Y also incorporates the 4 market scenarios. It shows a similar pattern as Fig X; On the right, when term slope becomes cumulatively more positive, the magnitude of ε_{T_N} increases, but at a decreasing rate. This holds for both up and down market states. But when the term slope becomes cumulatively more negative, the magnitude of ε_{T_N} increases at a faster rate. Furthermore, there is more curvature in the down-market state e.g. when cumulative term slope reaches -80% of the initial futures price, the 22% ε_{T_N} on downmarket is almost twice the magnitude of the upmarket's -12%. Hence Fig Y is consistent with Fig X in showing that ε_{T_N} becomes a serious concern during a downmarket under a cumulatively negative term slope⁶.

The graphs show that dispelling the roll-yield misconception requires more than a realization that rolling a long position on a cumulatively upward (downward) term slope, tend to understate (overstate) the return from a futures position. Our graphs show that, when term slope is cumulatively positive, if we directly aggregate a return series, CR_{jr} will overstate realized return from a loss position. But when returns are aggregated for a profit position, CR_{jr} will understate realized return. The same applies to a negative term slope; Figure X shows that whether CR_{jr} over or understates R_z , depends on whether R_z corresponds to a profit or loss position. Given that ε_{T_N} does not generate any actual profit/loss cashflow, it should be removed from CR_{jr} , so as to avoid misleading evaluation on the holding period return i.e. profitability offered by commodity futures as an asset class.

⁶Like Fig X, both graphs in Fig Y also cuts at the point of origin. This is because we make a simplifying assumption that under zero term slope, there are no jumps on every roll-day. If an investment horizon is described by a mixture of up and down jumps, it is possible for CJR_{T_i} to fluctuate around 1, but unlikely to be exactly 1. So ε_{T_N} can be empirically trivial, but not likely to be zero. Nonetheless, that doesn't change the nature of how ε_{T_N} behaves under different market scenarios.

	Positive slope $(1-CJR_{T_N}) > 0$	Negative slope $(1-CJR_{T_N}) < 0$
Up-market	$\varepsilon_{T_N} > 0$	$\varepsilon_{T_N} < 0$
Down-market	$\varepsilon_{T_N} < 0$	$\varepsilon_{T_N} > 0$
Effect of residual error	Under-estimate $ R_z $	Over-estimate $ R_z $

Table 1: Residual jump error under different market conditions.

$$\varepsilon_{T_N,t} = \phi_0 + (1 - CJR_{T_N,t})[\phi_1 + \phi_2 R_{z,t}] + \epsilon_t$$

$$\varepsilon_{T_N,t} = \phi_0 + (1 - CJR_{T_N,t})[\phi_1 + \phi_2 Dum_{R_z,t}] + \epsilon_t$$

$$, \text{ where } Dum_{R_z,t} \begin{cases} = 1, \text{ where } R_z < 0 \\ = 0, \text{ otherwise.} \end{cases}$$

$$|\Delta \varepsilon_{T_N,t}| = \phi_0 + (1 - CJR_{T_N,t})[\phi_1 + \phi_2 R_{z,t}] + \epsilon_t$$

$$|\Delta \varepsilon_{T_N,t}| = \phi_0 + (1 - CJR_{T_N,t})[\phi_1 + \phi_2 Dum_{R_z,t}] + \epsilon_t$$

Based on the table and eq(6), $E(\phi_1) > 0$ and $E(\phi_2) < 0$.

Hi Zihao, After discussing with Jari, we want to try running some regressions to complement his graphs. While the patterns in most of the graphs are consistent with Eq(6) and the simulation exercise, it is too monotonic to discuss every single one of them. I can discuss a few of the nicer pics. But what we need to finish off the paper, is a summary of the patterns across different markets and sample periods. The sample covers 12 commodities over 33 years. His sub-samples are 10+10+13 years.

The above are some regression specifications to try. The first 2 directly regress rje on the slope and interaction between slope and Rz level, and DumRz=1 for $R_z < 0$. The other 2 have similar specifications, but on rje volatility. Let's try running them on WTI crude and copper, for full sample and each of the 3 sub-samples. See what sort of results turn out. According to Eq(6), $E(\phi_1) > 0$ and $E(\phi_2) < 0$. Another issue to think about is, can we stack the commodities and run as a panel regression. The coefficients would be equivalent to average cross-sectional slopes over time, albeit on a small N=12.

To note, the main purpose of the numerical example is show readers how we compute different measures of cumulative return: $\{CR, CR_{jr}, R_z\}$. More importantly, to also visualize how ε_{T_N} could behave under each of the 4 market conditions. In particular, the numerical example is consistent with equation (6) in showing that ε_{T_N} is a potential concern if we

aggregate return during a downmarket under a negative term slope. In the next section, we shall verify this empirically using actual futures data on 12 different underlying commodities.

3.2 A numerical example

We focus our discussion on the \$0.50 jump in the top left quadrant of Table 1. A similar outline applies to the other three quadrants. In Panel A, the profit from each contract position is \$0.50, for a total of \$1.50. Under contango, the rolling from F1 to F2, and from F2 to F3, both occur an a \$0.50 up-jump \$10.50→\$11 and \$11.50→\$12. The sampled price series is $F_t = \{10, 10.5, 11.5, 12.5\}$. If we derive the return series R_t from F_t , and aggregate it, we obtain $CR = 25\%$. Clearly, this is wrong; Total profit from the 3 contract positions is \$1.50, or 15% when normalized by the initial price of \$10 that the position was established. If we adjust F_t using the JR method, generate the return series $R_{jr,t}$, we get $CR_{jr} = 14.35\%$. The latter is slightly downward biased, such that a 0.653% residual jump error remains.

The backwardation state further reinforces our reference to the discrepancy between R_z and CR_{jr} as an error. Except for flipping the term slope, all other details remain unchanged; The trader still makes \$1.50 from entering the initial futures position at \$10, such that $R_z = 15\%$. However, Table 1 shows that now, $CR = 5\%$ has been wrongly suppressed by down-jump errors in R_t . The CR_{jr} from the JR method is closer to R_z , but now its upward biased, which leads to $\varepsilon_{T_N} = -0.763\%$. To note, even though $R_{jr,t}$ is derived from same-contract prices, CR_{jr} is affected by whether the futures position is rolled along a positive (14.35%) or negative (15.76%) term slope. This happens even though the trader's profit position and entry price are the same in both states.

Next we examine the down-market scenario in the bottom left quadrant of Table 1; Now the trader has lost \$1.50 over 3 contract positions, and $R_z = -15\%$. In the contango state, unadjusted F_t gives $CR = -5\%$, which is upward biased due to up-jumps. For JR adjusted returns, $CR_{jr} = -14.26\%$ is closer to -15%, with a -0.737% residual jump error. But if the market is in backwardation, roll-day down-jumps exaggerate the unadjusted loss to $CR = -25\%$. But when we aggregate JR-adjusted return, $CR_{jr} = -15.89\%$ is closer to R_z , with $\varepsilon_{T_N} = 0.885\%$. Lastly, the right side of Table 1 corresponds to bigger roll-day jumps of \$1. The direction of ε_{T_N} is consistent with those under \$0.50 jumps, and they are around twice as large in magnitude. This is due to roll-day jump magnitudes that are twice as large.

Across the 8 scenarios, the realized return magnitude is $|15\%|$. Table 1 shows that, regardless of market direction, the CR from unadjusted F_t , deviates substantially from $|R_z|$. For a $\pm\$0.50$ jump, CR swings between 25% (contango) and 5% (backwardation) during up-market, and between between -5% (contango) and -25% (backwardation) during down-market. When the roll-day price jump doubles to $\pm\$1.00$, the swing range also doubles to $35\% \leftrightarrow -5\%$ for up-market, and $5\% \leftrightarrow -35\%$ for down-market. Based on $R_{jr,t}$, the fluctuation in CR_{jr} under different market conditions is substantially reduced; For a $\pm\$0.50$ jump, CR_{jr} oscillates between 14.35% and 15.76% for the up-market scenario, and between -14.26% to

-15.89% for the down-market scenario. And when the roll-day jump doubles to $\pm\$1$, the fluctuation in CR_{jr} also widens to 13.78% $\leftrightarrow$ 16.67% during up-market, and -13.64% $\leftrightarrow$ -16.97% for the down-market scenario.

In sum, Table 1 shows that although $R_{j,t}$ is free from jump error, its aggregation leaves a residual jump error in CR_{jr} that is a function of term slope. Within each of the four quadrants in Table 1, we hold the trader's profit/loss outcome and the initial price to set up the futures position, as constants. But CR_{jr} is affected by whether contango or backwardation prevails during the investment horizon.

The behavior in residual jump error ε_{T_N} under different market condition, is consistent with the predictions of equation (6). A rising market conforms with the roll yield misconception; During contango, $CR_{jr} = 14.35\% < 15\%$ is consistent with a long position generating a weaker return as it is being rolled by selling lower priced old contracts, and buying into higher priced new contracts. To follow, during backwardation, $CR_{jr} = 15.76\% > 15\%$ since a long position is being rolled by selling old contracts at higher prices, and buying back new contracts at lower prices. However, Table 1 also shows that the misconception is debunked in a falling market. During contango, $CR_{jr} = -14.26\%$ is greater than the actual loss of -15%. But during backwardation, $CR_{jr} = -15.89\%$ is smaller than the actual -15% loss. As such, regardless of market state, ε_{T_N} has opposite signs between up and down market direction. This is consistent with equation (6) showing that the sign of ε_{T_N} not only depends on how pervasive is the contango or backwardation state ($[1 - CJR_{T_i}]$), but also on market direction i.e. whether $R_z > 0$ or $R_z < 0$.

The values of CR_{jr} in Table 1 are also consistent with our intuition as to why would the sign of ε_{T_N} , depend on the interaction between market state and market direction. If contango dominates the investment horizon, F_t is adjusted using $CJR_{T_i} < 1$, which acts as a discount factor to 'pull down' a futures price series that has been inflated by roll-day up-jumps. This applies whether the market is rising or falling. The discounting effect causes CR_{jr} to understate the magnitude of R_z , *regardless* of its sign. For all four contango scenarios in Table 1, the magnitude of CR_{jr} is always less than 15%. Conversely, during backwardation, $CJR_{T_i} > 1$ acts as a compound factor to 'push up' F_t which has been periodically deflated by roll-day down-jumps. As such, CR_{jr} overstates the magnitude of R_z , regardless of actual market direction. In Table 1, all four backwardation scenarios correspond to CR_{jr} having a magnitude greater than 15%.

The focus will be on the five CR measures calculated from 3 price series: Bloomberg, SPDJ and an adjusted F_t series that is sampled directly from adjoining contracts, following the standard approach in the literature. We use CJR_{T_i} to adjust F_t so as to derive a return series from same-contract prices, which is free from jump errors. This is similar to how Erb and Harvey (2006) and Gorton and Rowenhurst (2006) compute futures return.

3.3 The rolling procedure

The empirical analysis is based on a daily sample that covers 12 commodity futures markets from 1989 to 2023. To demonstrate the relevant scope of residual jump error, we include different commodity types; 2 fuel (WTI crude, Gasoline), 3 metal (Gold, Silver, Copper), and 7 agricultural commodities (Cotton, Coffee, Sugar, Frozen Concentrated Orange Juice (FCOJ), Corn, Soybean, Wheat). We sample the Arabica Coffee C contract, which is more actively traded than the Robusta Coffee contract. Similarly, we sample the Chicago Soft Red Winter (SRW) Wheat contract, which has more trading volume than the KC-HRW contract. For Sugar, we sample from the Sugar #11 contract, which sets the global benchmark pricing of raw cane sugar. Lastly, the cotton sample refers to the flagship Cotton #2 contract.

We sample futures prices based on a pre-specified rolling procedure, following standard convention used by academics and analysts. This ensures that the residual jump error we document in this paper, is also relevant to them. There are various rules-of-thumb, most of which are aimed at ensuring that the sample is drawn from actively traded contracts over time, so that the futures price series resonates with trading activity.

To specify an appropriate roll-day for the rolling procedure, we locate days when there is a cross-over in open interest between the front and next-cycle contract. Take quarterly-cycle for example; Traders usually start rolling their positions around 10 to 14 days prior to the front contract’s expiry. When this occurs, open interest in the front contract starts to decline rapidly, which is accompanied by a rapid rise in the next-cycle contract’s open interest. For each commodity, we mark open-interest cross-over days over the full sample period, calculate the average cross-over day, and apply it as the ‘optimal’ roll-day T_i to sample from adjoining contracts over the full sample period⁷.

Contract cycles vary across commodity type; The agricultural commodities have between 4 to 7 contracts per year; WTI crude and Gasoline follow a monthly cycle, while the metals follow a quarterly cycle. The choice of T_i affects the magnitude of observed roll-day jump, but we argue that it is not necessary to consider a robustness check on T_i . First, our specified T_i is not ad-hoc; For each commodity, we use the cross-over in open-interest between adjoining futures contracts to identify each roll-day over the sample period, and use the commodity-specific average roll-day T_i to sample futures prices. This approach is empirically justified as it ensures our sampling procedure consistently tracks the contract with the greatest trading activity over time⁸. Second, the optimal T_i varies across commodities. This exposes our

⁷The optimal roll-day is affected by the first position day, which varies across commodities. This refers to the day when the Exchange can assign a delivery to a long futures position holder. For example Gold’s first position day is four weeks before contract maturity, whereas for WTI crude, it is the day after contract maturity. Only long hedgers who intent to take physical delivery would hold contracts pass the first position day. Furthermore, speculators have no interest in contracts that have passed this date. As such, rolling usually occurs a week before a commodity’s first position day. Usually the jump ratio is for the optimal roll-day, and it increases before and after the roll day. This applies to the front contract, but less so for the 2nd and 3rd cycle contract.

⁸In earlier analyses, we consider three ad-hoc roll-days: $(T - T_i) = \{\text{Close, Mid, Far}\}$, where Close = 2

empirical analysis to roll-day price jumps that covers a range of T_i values across commodities, which is equivalent to a robustness check on T_i . Third, our focus is on residual jump error, which is affected by the direction of cumulative price jump over time. Using a different T_i may have an impact on the magnitude of roll-day price jump, but it is very unlikely to change the direction of cumulative price jump over a long sample period.

3.4 Different cumulative return measures

We use the Bloomberg and SPDJ commodity price indices to generate daily return series for the 12 commodity futures markets. In addition, we sample daily settlement prices from adjoining contracts over time, based on the above rolling procedure. To address roll-day price jump, we adjust the sampled futures price series using a cumulative jump-ratio index CJR_{T_i} . We can confirm that our methodology produces a JR price series that is similar to how Bloomberg and SPDJ construct their commodity price indices; The JR price series generates a return series $R_{jr,t}$ that is based on same-contract prices. The return series is also consistent with how Erb and Harvey (2006) and Gordon and Rowenhurst (2006) construct futures return from same-contract prices.

Data vendors differ in their sampling and adjustment procedures, which could also vary across commodities. To ensure a consistent approach, we manually sample and adjust a futures price series to reduce potential noise in the cross-sectional analysis. More importantly, the return series derived from data vendor commodity price indices, do not allow us to pinpoint roll-days over time. As such, the proposed aggregation method is not readily applicable on Bloomberg and SPDJ return series. Instead of a pre-specified roll-day, data vendors apply a weighted-average rolling procedure to sample prices from adjoining contracts over a trading week⁹. Hence we cannot pin-point roll-days in the return series to implement the two-step aggregation. Second, data vendors' rolling procedures vary across commodity futures, which potentially complicates the cross-sectional analysis¹⁰. Third, during certain market or trading events, a data vendor would make ad-hoc adjustments to their price indices. But they do not elaborate what these adjustments are or how they are made. In contrast, our $R_{jr,t}$ is highly correlated with data vendor return series, but it also allows us to implement the proposed aggregation method.

Denote the Bloomberg, SPDJ and JR daily price series as $\{P_{Blg,t}, P_{Spdj,t}, P_{Jr,t}\}$. From

days to maturity, while Mid and Far depends on a commodity's contract cycle. For monthly cycle, we set Mid = 8 days and Far = 16 days to maturity; For quarterly cycle, we set Mid = 16 days and Far = 30 days to maturity. We find that roll-day choice does not impose a material effect on roll-day jumps. Subsequently, we use the open interest cross-over approach, which has a good justification, and allows T_i to vary across commodities.

⁹For example, Bloomberg applies a 5 trading-day window, where they gradually shift the weights from the old to new contract to compute the futures price.

¹⁰For example, the Bloomberg commodity price index rolls every two months because it samples from the 2nd-cycle contract; As it approaches expiry, the sampling rolls into the contract two cycles ahead; SPDJ samples from the front contract on a monthly roll.

each price series, we generate a discrete daily return series R_t , and track its cumulative return CR_t (running total) over the sample period. Denote $\{CR_{Blg}, CR_{Spdj}, CR_{Jr}\}$ as the cumulative return at the end of the sample period. These cumulative return measures are widely used by academics and industry analyst for back-testing and risk-return evaluation on futures positions. As we previously explain, the Bloomberg, SPDJ and JR daily return series $\{R_{Blg,t}, R_{Spdj,t}, R_{Jr,t}\}$ are derived from same-contract prices, such that $\{CR_{Blg}, CR_{Spdj}, CR_{Jr}\}$ all contain residual jump error.

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \quad CR_t = \prod_{i=1}^t (1 + R_i) - 1, \text{ where } t = 1, 2, \dots, T$$

Unlike $R_{Blg,t}$ and $R_{Spdj,t}$, we can pinpoint every roll-day T_i for $R_{Jr,t}$ over the sample period. This allows us to apply Step 2A and 2B to generate two additional cumulative return series $CR_{A,t}$ and $CR_{B,t}$. Below we outline Step 2A to generate $CR_{A,t}$. When $t \neq T_i + 1$, we can aggregate same-contract returns as per normal. But for every $t = T_i + 1, i = 1, 2, \dots, N$, the cumulative return will cross adjoining contracts. Step 2A takes a linear sum between the cumulative return CR_{A,T_i} at T_i , and R_{Jr,T_i+1} i.e. the first return from the newly rolled contract at $T_i + 1$. After this, CR_{A,T_i+1} continues on to aggregate same-contract returns from the newly rolled position as per normal, until it crosses the next roll-day, and so forth.

$$CR_{A,t} = \begin{cases} \prod_{i=1}^t (1 + R_{Jr,i}) - 1, & t = 1, 2, \dots, \text{ for } t \neq T_i + 1, \\ CR_{A,T_i} + R_{Jr,T_i+1}, & \text{ for } t = T_i + 1, \forall i = 1, 2, \dots, N. \end{cases}$$

A similar aggregation method applies to generate $CR_{B,t}$, albeit a more complex adjustment is required in Step 2B, compare to Step 2A. The outline below follows from the detailed discussion on Step 2B in equation (4). Step 1 is the same; Same contract returns correspond to an on-going position, so they can be aggregated as per normal for $t \neq T_i + 1$. But when $t = T_i + 1$, the cumulative return crosses from old to new contract. Here, Step 2B applies a scaling on $(1 + R_{Jr,T_i+1})$, the first return of the newly rolled contract position, before taking its product-sum with CR_{B,T_i} to get CR_{B,T_i+1} . After that, the usual aggregation applies to the same-contract return series, until the next roll-day, and so forth.

$$CR_{B,t} = \begin{cases} \prod_{i=1}^t (1 + R_{Jr,i}) - 1, & t = 1, 2, \dots, \text{ for } t \neq T_i + 1, \\ CR_{B,T_i} \times \frac{F_{i,T_i-1}}{F_{i-1,T_{i-1}} + \sum_{j=1}^{N-2} J_{T_j}} (1 + R_{Jr,T_i+1}) - 1, & \text{ for } t = T_i + 1, \forall i = 1, 2, \dots, N. \end{cases}$$

In sum, we have 5 cumulative daily return series $\{CR_{Blg,t}, CR_{Spdj,t}, CR_{Jr,t}; CR_{A,t}, CR_{B,t}\}$, which we test against market state variables and evaluate against the realized return R_z for the 12 commodities. The first 3 series $\{CR_{Blg,t}, CR_{Spdj,t}, CR_{Jr,t}\}$ come from directly aggregating returns over time, which causes residual jump error. Equations (3) and (4) show that $CR_{A,t}$ and $CR_{B,t}$ do not contain residual jump error. But given that Step 2A is a quick but inaccurate way to aggregate cross-contract returns over time, $CR_{A,t}$ may exhibit some unknown approximation error.

INSERT TABLE 1 HERE

In Table 1, we report the cumulative return $\{CR_{Blg}, CR_{Spdj}, CR_{jr}\}$ for the 12 commodities over the full sample period. It shows that the 3 return series yield consistent cumulative returns in 6 commodities: Gold, Cotton, Sugar, Coffee, Corn and Wheat. The JR cumulative return differs from the two data vendors for Silver, FCOJ and Soybean. For example, The cumulative return for Silver is 61.2% according to CR_{Blg} and CR_{Spdj} , but it is 47.2% from CR_{jr} . Lastly, for WTI, Gasoline and Copper, the three cumulative return measures differ from one another e.g. For Gasoline, $CR_{Blg}=555.64\%$, $CR_{Spdj}=824.64\%$ and $CR_{jr}=1102.27\%$. To note, any discrepancy in cumulative return is generated over 30 years; If we had reported annualized return, any discrepancy in return would be even smaller. Furthermore, due to differences in sampling choices among the 3 methods e.g. rolling-procedure, ad-hoc adjustments by data vendors etc, we expect there to be some discrepancies among the different cumulative return measures. The main point shown in Table 1 is that the JR method is a comparable approach to how data vendors adjust the raw futures price series to drive a return series from same-contract prices.

Present and discuss correlation matrix for cumulative return

We empirically measure residual jump error in equation (6) as the difference between the various measures of cumulative return CR_t and realized return $R_{z,t}$ over time. For the latter, we track the cumulative daily profit/loss cashflow $\sum CF_t$ over time, assuming the futures investor switches contract positions on the same roll-day T_i as the rolling procedure. At every $t = T_i$, we manually check to ensure that $CF_{T_i} = F_{i,T_i} - F_{i,T_i-1}$ is derived from settlement prices of the expiring contract. To follow, at every $t = T_i + 1$, we make sure that $CF_{T_i+1} = F_{i+1,T_i+1} - F_{i+1,T_i}$ is the settlement price change of the newly rolled contract $i + 1$. The $\sum CF_t$ series represents the cumulative marked-to-market profit/loss on a futures trader's margin account, which we normalize by the initial futures price F_{1,T_0} to compute $R_{z,t}$, and generate residual jump errors for the 5 CR_t measures in Table 2, denote as $\varepsilon_{k,t}$, where $k = \{Blg, Spdj, Jr, A, B\}$.

INSERT TABLE 2 HERE

3.5 Residual jump error over time

We test if $\varepsilon_{k,t}$ fluctuate over time with variables that proxy market state and roll-yield. If a correct aggregation method is applied, the cumulative return series should not contain residual jump error, and $\varepsilon_{k,t}$ should not co-move with the proxy measures over time.

But an erroneous return aggregation will cause CR_t to be a function of term slope, causing it to co-moves with proxy measures of market state and roll-yield over time.

To note, all 5 cumulative return series correspond to a buy-and-hold position. Hence the proxy measures of market state and roll-yield, cannot contemporaneously affect cumulative return over time. If we are using roll-yield or term slope as part of a futures trading strategy e.g. to time the opening and closing of a position, then lagged market state and roll-yield variables could explain cumulative return over time.

We construct two measures in equation (7); First is the cumulative net Up-jump CUp_t as a proportion of total roll-days N , where $Up_{T_i} = 1$ for a roll-day up-jump, and $Down_{T_i} = 1$ for a roll-day down-jump. To note, CUp_t changes only on roll-days, and acts like a step-wise dummy that tracks the pervasiveness of market state. For a market that is persistently in contango, its CUp_t will gradually drift towards 1 over time. In contrast, for a market that periodically switches between contango and backwardation, CUp_t will fluctuate around zero.

The second measure is the cumulative roll-yield CRy_t ; Following the literature, we calculate roll-yield as $Ry_t = \frac{F_{T_i}^n - F_{T_i}^o}{F_{T_i}^o}$ on roll-days $t = T_i$, and measure $CRy_t = \prod_t^T (1 + Ry_t) - 1$. On non roll-days, $Ry_t = 0$, such that CRy_t changes only on roll-days. We could also view CRy_t as another measure of the persistence in market state. If a market is consistently in contango, we are likely to observe $Ry_t > 0$ over roll-days, and CRy_t will gradually increase over time. But if a market periodically switches between contango and backwardation, the Ry_t series will exhibit a good mix of positive and negative yields, such that CRy_t fluctuates around zero.

$$CUp_t = \sum_t^T [Up_t - Down_t]/N,$$

where $Up_{t=T_i} = 1$ for up-jump; $Down_{t=T_i} = 1$ for down-jump

$$Ry_t = \begin{cases} \frac{F_{T_i}^n - F_{T_i}^o}{F_{T_i}^o}, & \text{for } t = T_i \\ 0, & \text{otherwise.} \end{cases} \quad CRy_t = \prod_t^T (1 + Ry_t) - 1 \quad (7)$$

For each commodity market, we regress each of the $k = \{Blg, Spdj, Jr, A, B\}$ cumulative return series $CR_{k,t}$ on CUp_t and CRy_t over the full sample period.

$$CR_{k,t} = \phi_{k,0} + \phi_{k,1}CUp_t + \phi_{k,2}CRy_t + \epsilon_{k,t} \quad (8)$$

We hypothesize $\phi_{1,k}, \phi_{2,k}$ to be significantly negative for $k = \{Blg, Spdj, Jr\}$. When CUp_t and CRy_t trend upwards over the sample period, this corresponds to a persistently contango market. Our analytical representation of residual jump error shows that a contango market will generate $\varepsilon_{T_i} < 0$. Take CRy_t for example; In a contango market, CRy_t as a measure is positive and increasing. But its impact on return is negative. A long investor who rolls along a positive term slope, is periodically having to close expiring positions at lower prices, and buy into higher priced contracts. This drags down $CR_{k,t}$, such that $\phi_{2,k} < 0$. And vice versa during a backwardation market. To follow, we hypothesize that for $k = \{A, B\}$, $\phi_{1,k}, \phi_{2,k}$ are insignificant. Earlier, we show that by using Step 2A or 2B, the aggregated JR return is no longer a function of term slope. Hence we do not expect $CR_{A,t}$ and $CR_{B,t}$ to co-move with CUp_t and CRy_t over time.

In addition to running separate time-series regressions for each of the 12 commodities, we could instead stack the time-series regressions as a 12-market panel, and run panel regression for each CR measure to test the significance of $\phi_{1,k}$ and $\phi_{2,k}$, which averages across the 12 markets. Given we draw from different commodity types, its more appropriate to use a time-FE specification for the panel regression.

Given the residual jump error problem is caused by aggregating futures return across contracts, it seems natural to analyze time-series effect of market state on return in contract-cycle frequency, rather than daily frequency. This is awkward to implement empirically, as the different commodity types follow different contract cycles; WTI and Gasoline follow monthly cycles; The agricultural commodities have 5 contracts per year, which are not consistently spread out across commodities. Lastly, the metals Gold, Silver and Copper, follow quarterly contract cycles. By analyzing the time-series effects in daily frequency, it provides us flexibility on how frequent we want to examine the cross-sectional slope of CR on market state measures. We elaborate in the next section.

3.6 Cross-sectional effects in residual jump error

If the pervasiveness of market state affects the direction and severity of residual jump error in cumulative return, this implies a cross-sectional pattern; Contango (Backwardation) markets exhibit negative (positive) residual jump error in cumulative return, while markets that oscillate between contango and backwardation, should have less severe residual jump error.

From the daily variables $CR_{k,t}, CUp_t, CRy_t$, we extract the end-of-month observations across $j = 1, 2, \dots, 12$ commodities. For each of the five CR measures, we run monthly cross-sectional regressions in equation (9). Each monthly regression provides a snapshot of the cross-sectional slope of $CR_{k,j}$ against each of the two market state variables. For each $k =$

$\{Blg, Spdj, Jr\}$ over the sample period, the monthly regression generates a time-series of around 12 x 30 years = 360 estimated cross-sectional slopes $\lambda_{1,t_m} = \{\lambda_{1,t_1}, \lambda_{1,t_2}, \dots, \lambda_{1,T_m}\}$ for CUP_j , and $\lambda_{2,t_m} = \{\lambda_{2,t_1}, \dots, \lambda_{2,T_m}\}$ for CRy_j . We calculate the average slopes $\bar{\lambda}_1, \bar{\lambda}_2$ as well as the t-stats to ascertain if the average cross-sectional slopes' magnitude is non-trivial and stable over time.

$$CR_{k,j} = \lambda_{k,0} + \lambda_{k,1}CUP_j + \lambda_{k,2}CRy_j + u_{k,j}$$

$$\bar{\lambda} = \frac{\sum_{t_m}^{T_m} (\lambda_{t_m})}{T_m}; \quad \sigma_{\lambda_{t_m}}^2 = \frac{\sum_{t_m}^{T_m} (\lambda_{t_m} - \bar{\lambda})^2}{T_m - 1}$$

$$\text{t-stat} = \frac{\bar{\lambda}}{\sigma_{\lambda_{t_m}}} \quad (9)$$

For $k = \{Blg, Spdj, Jr\}$, we show analytically that $CR_{k,j}$ is a function of term slope. Commodity markets that are in contango (backwardation) have larger (smaller) values for CUP_j and CRp_j , as well as negative (positive) residual jump error in cumulative return. If this effect is empirically evident, it will manifest as significantly negative cross-sectional average slopes $\bar{\lambda}_{k,1} < 0$ and $\bar{\lambda}_{k,2} < 0$. In contrast, since $CR_{A,t}$ and $CR_{B,t}$ are not functions of term slope, there should not be a cross-sectional relation between cumulative return under Step 2A and 2B, and the market state variables i.e. $(\bar{\lambda}_{A,1}, \bar{\lambda}_{A,2})$ and $(\bar{\lambda}_{B,1}, \bar{\lambda}_{B,2})$ should all be insignificant.

4 Conclusion

Hi Jari,

Maybe we focus our discussion on WTI? After we are satisfied with the coverage, you can then map onto the other 11 commodities.

The graphs and the table of CR measures are complements; They need to be presented together.

The 3 standard CR measures are based on return series from Bloomberg, SPDJ and JR method.

The 2 proposed CR measures CRA and CRB are based on the JR return series. They require us to pinpoint roll-days, hence not applicable to data vendor return series.

Rz is a single number that is represented as a dot in the graph.

BASELINE RESULT: Full sample period over 30 years

We plot the 5 CR measures over the full sample period. Here we don't highlight market state or direction. We just give a snapshot of how the 5 CR series vary over a long sample period, and where they end up at the end of the full sample period, compare to R_z .

MAIN RESULTS: Sub-samples partitioned by market state and market direction.

Time-series 1: Fix commodity and market state;

Compare residual jump error between up-market vs. down-market period.

Time-series 2: Fix commodity and market direction;

Compare residual jump error between C and B. To note, this applies to a limited number of commodities, as states are pervasive in the majority of several commodities. But I think WTI is 1st-half (15 years) in one state, and the 2nd-half in the other state.

Cross-section 1: Fix market direction;

Compare residual jump error between C and B commodities.

Cross-section 2: Fix market state;

Compare residual jump error between up-market and down-market commodities.

Regardless of which 4 states comparison, CRA should always end up close to R_z , and CRB should always end up equal to R_z .